

Tyler J. Law, MD, MSc

University of California, San Francisco, San Francisco, CA ▪ tyler.law@ucsf.edu —

EDUCATION

Anesthesia Critical Care Medicine Fellowship University of California, San Francisco, Department of Anesthesia and Perioperative Care	2021
Global Health Equity Fellowship University of California, San Francisco, Department of Anesthesia and Perioperative Care	2018
Fellow of the Royal College of Physicians and Surgeons (Anesthesiology) University of Toronto, Toronto, Canada	2017
Master of Science, Health Policy, Planning and Financing London School of Economics, London, UK	2015
Doctor of Medicine Queen's University, Kingston, Canada	2012
Bachelor of Health Sciences McMaster University, Hamilton, Canada	2008

PRINCIPAL APPOINTMENTS

Associate Professor University of California, San Francisco, Department of Anesthesiology & Perioperative Care	July, 2025 – present
Assistant Professor University of California, San Francisco, Department of Anesthesiology & Perioperative Care	July, 2018 – July, 2025
Clinical Instructor (Anesthesiology) University of California, San Francisco	July, 2017 – June, 2018
Attending Physician (Anesthesiology) Zuckerberg San Francisco General Hospital	July, 2017 – June, 2018

OTHER POSITIONS

Chair, Code Blue Committee Zuckerberg San Francisco General Hospital	2022 - current
Associate Director of Research, Center for Health Equity in Surgery and Anesthesia University of California, San Francisco	2022 - current
Faculty Recruitment Committee, ZSFG	2022 - current
Research Assistant, LSE Health London School of Economics, London, UK	January, 2015 – September, 2015
Visiting Research Student, Outcomes After Critical Illness and Surgery Johns Hopkins University, Baltimore, USA	May – August, 2010
Research Fellow, McMaster University Health Forum McMaster University, Hamilton, ON	2009
Intern, National Institute for Health Research, Department of Health London, United Kingdom	August, 2008
Intern, Alliance for Health Policy and Systems Research World Health Organization, Geneva, Switzerland	April – July, 2008

RESEARCH

PEER-REVIEWED PUBLICATIONS

1. Hughes C, Chen D, **Law TJ**, et al. Pulse oximeter performance and skin pigment: Comparison of 34 oximeters using current and emerging regulatory frameworks. *Anesth Analg* 2026;10.
2. Corpman D, Lang E, **Law TJ**. Large language models in trauma anesthesia education. *Curr Opin Anaesthesiol* 2026;39(2):160–166.
3. Dundas NE, **Law T**, Brender T, et al. All that shines is not gold: Maintaining scientific rigor when evaluating, interpreting, and reviewing studies using large language models. *Anesthesiology* 2026;144(2):272–288.
4. **Law TJ**, Leeb G, van Wijk XMR, et al. Differences in arterial oxygen saturation measurements between two co-oximeters from different manufacturers. *Respir Care* 2025;70(12):1485–1493.
5. Elmankabadi S, Dove J, Behnke E, et al. Comparing pulse oximeter performance using a common functional tester versus controlled desaturation studies on healthy participants. *J Clin Monit Comput* 2025;1–12.
6. Fong N, Lipnick MS, Behnke E, Chou Y, Elmankabadi S, Ortiz L, et al. Open access dataset and common data model for pulse oximeter performance data. *Sci Data* 2025;12(1):570.
7. Nankabirwa V, Guma I, Kabunga J, Bigogo HC, Navume DE, Mwanje TK, **Law TJ**, et al. Acute trauma care training in undergraduate medical education programs in Uganda; A cross-sectional survey of final year medical students. *Afr J Emerg Med* 2025;15(2):602–606.
8. Leir, Sarah A., **Tyler J. Law**, and M. Dylan Bould. 2024. "The Anesthesia Human Resources Crisis in Canada." *Journal Canadien d'anesthésie [Canadian Journal of Anaesthesia]* 71 (12): 1612–26.
9. Miclau, Theodore A., Daniel J. Bennett, Laura Lang, Megha Parekh, and **Tyler J. Law**. 2024. "Rideshare-Based Transportation Services for Surgical Patients in an Urban Setting: A Descriptive Analysis of a Quality Initiative Program." *Journal of Transport & Health* 36 (May): 101813.
10. **Law, Tyler J.**, Michael S. Lipnick, Wayne Morriss, Adrian W. Gelb, Jannicke Mellin-Olsen, Daniela Filipescu, Jackie Rowles, et al. 2024. "The Global Anesthesia Workforce Survey: Updates and Trends in the Anesthesia Workforce." *Anesthesia & Analgesia*, March, 10.1213/ANE.0000000000006836.
11. Leeb, Gregory, Isabella Auchus, **Tyler Law**, Philip Bickler, John Feiner, Shamsudini Hashi, Ellis Monk, et al. 2024. "The Performance of 11 Fingertip Pulse Oximeters during Hypoxemia in Healthy Human Participants with Varied, Quantified Skin Pigment." *EBioMedicine* 102 (March): 105051.
12. Petrucci, Beatriz, Samuel Okerosi, Rolvix H. Patterson, Sara B. Hobday, Valerie Salano, Christopher J. Waterworth, **Tyler J. Law**, Robert M. Brody, et al. 2023. "The Global Otolaryngology-Head and Neck Surgery Workforce." *JAMA Otolaryngology– Head & Neck Surgery*, August. <https://doi.org/10.1001/jamaoto.2023.2339>.
13. Lilaonitkul, M., A. Zacharia, **T. J. Law**, N. Yusuf, P. Saria, and J. Moore. 2023. "Evaluation of Practice Change Following SAFE Obstetric Courses in Tanzania: A Prospective Cohort Study." *Anaesthesia*, July. <https://doi.org/10.1111/anae.16091>.
14. Fong, Nicholas, Erica Langnas, **Tyler Law**, Mallika Reddy, Michael Lipnick, and Romain Pirracchio. 2023. "Availability of Information Needed to Evaluate Algorithmic Fairness - A Systematic Review of Publicly Accessible Critical Care Databases." *Anaesthesia, Critical Care & Pain Medicine*, May, 101248.
15. Percy, Samuel, Saad Sahi, Emmanuel Bua, Sriram Shamasunder, Michael Lipnick, and **Tyler Law**. 2023. "Living the Work: The HEAL Initiative as a Model for Perioperative Health Workforce Transformation and Health Equity Work." *Canadian Journal of Anaesthesia = Journal Canadien D'anesthésie*, May, 1–4.
16. **Law, Tyler J.**, Derek Stephens, and James G. Wright. 2022. "Surgical Wait Times and Socioeconomic Status in a Public Healthcare System: A Retrospective Analysis." *BMC Health Services Research* 22 (1): 579.
17. Henson, Laurence Cyril O., Courtney R. Lyles, Christy J. Khouderchah, and **Tyler J. Law**. 2021. "Impact of Gentrification on Asthma Visits to the Emergency Department." *SN Social Sciences* 2 (1): 3.
18. **Law, Tyler J.**, Shivani Subhedar, Fred Bulamba, Nathan N. O'Hara, Mary T. Nabukenya, Cornelius Sendagire, Adam Hewitt-Smith, Michael S. Lipnick, and Janat Tumukunde. 2021. "Factors Affecting Job Choice among Physician Anesthesia Providers in Uganda: A Survey of Income Composition, Discrete Choice Experiment, and Implications for the Decision to Work Rurally." *Human Resources for Health* 19 (1): 93.
19. Sund, G. C., M. S. Lipnick, **T. J. Law**, E. A. Wollner, and G. E. Rwibuka. 2021. "Anaesthesia Facility Evaluation: A Whatsapp Survey of Hospitals in Burundi." *South African Journal of Anaesthesia and Analgesia* 27 (2). <http://www.sajaa.co.za/index.php/sajaa/article/view/2456>.
20. Wollner, Elliot, Maziar M. Nourian, William Booth, Sophia Conover, **Tyler Law**, Maytinee Lilaonitkul, Adrian W. Gelb, and Michael S. Lipnick. 2020. "Impact of Capnography on Patient Safety in High- and Low-Income Settings: A Scoping Review." *British Journal of Anaesthesia* 125 (1): e88–103.
21. Wollner, Elliot, **Tyler Law**, Kristina Sullivan, and Michael S. Lipnick. 2020. "Why Every Anesthesia Trainee Should Receive Global Health Equity Education." *Canadian Journal of Anaesthesia = Journal Canadien D'anesthésie* 67 (8): 924–35.
22. Zha, Yuanting, Paul Truché, Erick Izquierdo, Kathrin Zimmerman, Sandra de Izquierdo, Michael S. Lipnick, **Tyler J. Law**, Adrian W. Gelb, and Faye M. Evans. 2020. "Assessment of Anesthesia Capacity in Public

Surgical Hospitals in Guatemala.” Anesthesia and Analgesia, November. <https://doi.org/10.1213/ANE.0000000000005297>.

23. **Law, Tyler J.**, Michael Lipnick, Muralidhar Joshi, Girija P. Rath, and Adrian W. Gelb. 2019. “The Path to Safe and Accessible Anaesthesia Care.” Indian Journal of Anaesthesia 63 (12): 965–71.
24. **Law, Tyler J.**, Fred Bulamba, John Paul Ochieng, Hilary Edgcombe, Victoria Thwaites, Adam Hewitt-Smith, Eugene Zoumenou, et al. 2019. “Anesthesia Provider Training and Practice Models: A Survey of Africa.” Anesthesia & Analgesia 129 (3): 839.
25. Bulamba, Fred, Cornelius Sendagire, Andrew Kintu, Adam Hewitt-Smith, Fred Musana, Maytinee Lilaonitkul, Emmanuel T. Ayebele, **Tyler Law**, et al. 2018. “Feasibility of Simulation-Based Medical Education in a Low-Income Country: Challenges and Solutions From a 3-Year Pilot Program in Uganda.” Simulation in Healthcare: Journal of the Society for Simulation in Healthcare. <https://europepmc.org/abstract/med/30601468>.
26. Salcher M, Naci H, **Law TJ**, Kuehne T, Schubert S, Kelm M, et al. Balloon Dilatation and Stenting for Aortic Coarctation: A Systematic Review and Meta-Analysis. Circ Cardiovasc Interv. 2016;9(6).
27. **Law TJ**, Leistikow NA, Hoofring L, Krumm SK, Neufeld KJ, Needham DM. A survey of nurses’ perceptions of the intensive care delirium screening checklist. Dynamics. 2012 Winter;23(4):18-24.
28. **Law TJ**, Lavis JN, Hamandi A, Cheung A, El-Jardali F, KT Platform Evaluation Team. Climate for evidence-informed health systems: A profile of systematic review production in 41 low- and middle-income countries, 1996-2008. J Health Serv Res Policy. 2012 Jan;17(1):4-10.
29. Desai SV, **Law TJ**, Needham DM. Long-term Complications of Critical Care. Crit Care Med. 2011 Feb;39(2):371-9.
30. Ranson K, **Law TJ**, Bennett S. Establishing health systems financing research priorities in developing countries using a participatory methodology. Soc Sci Med 2010 Jun;70(12):1933-42.

OTHER PUBLICATIONS

1. Rose, John, and **Law, Tyler**. 2023. “The Madagascar Experience: A Step Forward in Population-Level Evidence to Guide National Surgical Obstetrics and Anesthesia Planning.” Canadian Journal of Anaesthesia = Journal Canadien D’anesthesie.
2. **Law, Tyler J.**, John Rose, and Adrian W. Gelb. 2022. “Initiatives to Support Rural Access to Anesthesia.” Canadian Journal of Anaesthesia.
3. Chapter 2: Anesthesia in the Global Context (Canada section). **Law, TJ**. In: Miller’s Anesthesia, 9th edition. Ed: Gropper, M. Miller, R. 2019.
4. Avoidable Surgeries: Description and Identification of a Novel Health Systems Indicator. **Law, TJ**. London School of Economics. 2015. (Master’s Thesis).
5. Desai SV, **Law TJ**, Bienvenu OJ, Needham DM. Response to Psychiatric long-term complications of intensive care unit survivors. Crit Care Med. 2012.
6. AFMC Primer on Population Health. Reviewer for Chapter 12: The Organisation of Health Services in Canada, and Chapter 14: Decision-making: Policies and Ethics in Health Care and Public Health. http://phprimer.afmc.ca/inner/primer_contents. 2011.

CURRENT AND COMPLETED RESEARCH FUNDING

ONGOING RESEARCH SUPPORT

International Anesthesia Research Society	Law (PI)	08/2025 - 08/2027
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Funding amount: \$175,000 for FTE
 “Prediction of Pulse Oximeter Bias Using Machine Learning”
Role: Principal Investigator (data analytics - developing machine learning models to predict pulse oximeter bias based on skin tone, perfusion, and other factors)

FDA CERSI - Grant#70	Bickler, Hendrickson, Lipnick (PIs)	09/2023 - 08/2024
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Funding amount: \$23,431 for FTE
 “Prospective Clinical Study of Pulse Oximeter Errors in Hospitalized Patients.”
 The “Equiox” clinical study of pulse oximeters and skin tone in critically ill patients
Role: Key personnel (data analytics – dashboard reporting metrics, descriptive analysis, quality metrics, outlier investigation.)

UNITAID/PATH	Lipnick (PI)	03/2023 - 06/2024
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Funding amount: \$50,534 for FTE
 Open Oximetry: Create and manage an open access oximeter performance data repository for pulse oximeters, including raw signal data, perfusion and skin pigment data, populated by our

lab and other teams using standardized data collection methods established by the Open Oximetry project.

Role: Key personnel (data analytics - managing data engineering pipeline, database structure, data cleaning, QI, and project support)

Completed Research Support

USAID Assessments	Lipnick (PI)	10/2022 - 03/2024
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Funding amount: \$30,064 for FTE

Assessing USAID funded implementation of oxygen infrastructure (LOX, PSA) support projects, and Test-to-Treat uptake implementation.

Role: Key personnel (planning and implementing data capture processes, survey implementation and analysis)

CHESA Seed Award	Law (PI)	08/2022 - 08/2023
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Funding amount: \$2000

Understanding anesthesia graduates' practice environments and reach to underserved communities

USAID COVID-19 (Global Goods)	Lipnick (PI)	12/2021 - 11/2023
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Funding amount: \$101,570 for FTE

Working with USAID and USAID's NGO Implementing Partners to create global goods for supporting country level teams with COVID19 clinical care and respiratory care.

Role: Key personnel (supporting creation of COVID19 related decision aids, educational tools, webinars etc.)

USAID COVID-19 (02)	Lipnick (PI)	05/2020 - 12/2021
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Funding amount: \$45,243 for FTE

Continuing the work from the previous award to support technical teams in the COVID response and create clinical case management tools for global consumption.

Role: Key personnel (supporting creation of COVID19 related decision aids, educational tools, webinars etc.)

PRESENTATIONS

1. **Challenges in Global Surgical Workforce Assessment** World Congress of Anaesthesia, Marrakech, Morocco, 2026 [↗](#)
2. **Anesthesia Workforce Measurement - Multidisciplinary Lessons Learned** World Congress of Anaesthesia, Marrakech, Morocco, 2026 [↗](#)
3. **AI for the Medical and Health Researcher.** Institute of Global Health Sciences, San Francisco, May 2026 [↗](#)
4. **Code Blues - From Algorithm to Physiology** ZSFG RN Education Day, 2026
5. **Comparison of Objective and Subjective Tools for Assessing Skin Pigmentation Diversity.** SPIE Photonics West, San Francisco, 2025
6. **The Human Resource Factors in Anesthesia Safety.** Fighting Fatigue Together: Prioritizing Workforce Well-being for Enhanced Patient Safety (European Patient Safety Foundation webinar), June 12, 2024
7. **Mapping the Global Anesthesia Workforce.** European Society of Anesthesia and Intensive Care Medicine (ESAIC) Conference, Munich, Germany, 2024
8. **Challenges in Workforce Data Collection.** Utstein Business Meeting, World Congress of Anesthesia Conference, Singapore, 2024.

9. **The WFSA Workforce Survey.** International Anesthesia Research Society. Denver, CO. 2023.
10. **Pulse oximetry, health disparities & opportunities for data sharing.** Anesthesia Critical Care Speaker Exchange Program. October, 2023.
11. **The WFSA Workforce Survey: the final data.** American Society of Anesthesiologists 2022 Conference, New Orleans, USA, October 24, 2022
12. **Counting and Retaining the Anesthesia Workforce.** All Africa Anesthesia Conference. Kigali, Rwanda. September 10, 2022.
13. **Anesthesia, Critical Care, and UHC.** 13th International Society for Priorities in Health Conference. Bergen, Norway (virtual). April 28, 2022.
14. **Challenges of Assessing Oxygen Use, and the Clinical Care Tools Developed during the Pandemic.** 22nd ASEAN Congress of Anesthesiologists, Hanoi, Vietnam (virtual). March 19, 2022.
15. **The Global Anesthesia Workforce: Considerations for Measurement** Global OHNS Initiative. June, 2020.
16. **Surgical wait times and socioeconomic status in a publicly funded health care system.** Tyler J. Law, Derek Stephens, James G. Wright. American Society of Anesthesiologists 2018 Conference, San Francisco, California
17. **National Workforce Planning - A model.** Association of Anesthesiologists of Uganda Conference, Kampala, Uganda, 2018
18. **Student superheroes in interprofessionalism.** National Health Science Students' Association Conference, Toronto, 2011
19. **Interprofessionalism: Who, when?** Virginia Tech Carilion School of Medicine, Roanoke, Virginia, 2010

Abstracts

Examination of sex-linked differences in resting SpO₂ and SaO₂ in healthy adults. Yu (Celine) Chou, Danni Chen, Ella Behnke, Lily Ortiz, René Vargas Zamora, John Feiner, Tyler Law, Philip Bickler, Carolyn Hendrickson, Fekir Negussie, Michael S. Lipnick. American Thoracic Society, San Francisco, CA, 2025

Investigating Intrinsic Bias in Publically Available Critical Care Datasets for Machine Learning. Fong, Nicholas, Erica Langnas, Tyler Law, Mallika Reddy, Michael Lipnick, and Romain Pirracchio. International Anesthesia Research Society, Denver, CO, 2023

Rideshare-Based Transportation Services for Pre- and Post-Operative Surgical patients in an Urban Setting: A Descriptive Analysis of a Quality Initiative Program. Miclau, Theodore A., Daniel J. Bennett, Laura Lang, Megha Parekh, and Tyler Law. UCSF FCM Colloquium, San Francisco, CA, 2022.

Heterogeneity of non-physician anesthesia provider training and practice models: a survey of Africa. Tyler J. Law, MD, MSc, Fred Bulamba MBChB, J.P. Ochieng, MBChB4, Hilary Edgecombe, MBBS2, Victoria Thwaites, MBBS3, A. Hewitt Smith, MBBS, Eugene Zoumenou, Maytinee Lilaonitkul, MBBS, Adrian Gelb, MD, Redeit Shimeles, Paulin Banguti, Dylan Bould, Pascal Rod, Jackie Rowles, Francisco Lobo, Michael S. Lipnick, MD The Global Anesthesia Workforce Study Group Stanford-UCSF Anesthesia Research Retreat, San Francisco, 2018 UCSF Research Day, San Francisco, 2018 American Society of Anesthesiologists, San Francisco, 2018

Heterogeneity in anesthesia practice models worldwide: a descriptive review of 11 high-income countries. Aya Abounasr, Michael S. Lipnick, Adrian Gelb, Elise A. Delagnes, Maytinee Lilaonitkul, Tyler Law. American Society of Anesthesiologists, San Francisco, 2018 UCSF Research Day, San Francisco, 2018

Surgical wait times and socioeconomic status in a publicly funded health care system.
Tyler J. Law, Derek Stephens, James G. Wright. American Society of Anesthesiologists, San Francisco, 2018

Development Of A Facility-based, Anesthesia Capacity Assessment Tool For Resource-constrained Settings. Elise Delagnes, M.D, M.A, Shirley Zhao, B.S., Fred Bulamba, M.D., **Tyler J. Law,** M.D., Adrian W. Gelb, M.D., Stephen Ttendo, M.D., Adam Hewitt-Smith, M.D., Joseph Kiwanuka, M.D., Faye M. Evans, M.D., Michael S. Lipnick, M.D. American Society of Anesthesiologists, San Francisco, 2018 UCSF Research Day, San Francisco, 2018 College of Surgeons of East, Central, and South Africa, Kigali, Rwanda, 2018

AWARDS AND SCHOLARSHIPS

First Place, Editor’s Choice Category, Anesthesia & Analgesia March, 2025
For the paper: “The Global Anesthesia Workforce Survey: Updates and Trends in the Anesthesia Workforce”

Early Career Global Health Scientist Award, Institute for Global Health Sciences, UCSF
<https://globalhealthsciences.ucsf.edu/early-career-global-health-scientist-awards-highlight-projects-that-reimagine-global-health/> Jan, 2024

Academy of Educators Excellence in Teaching Award, UCSF School of Medicine Sept, 2023

STUDENTS MENTORED

- 1. **Aya Abounasr (2017-2018):** Mentored 4th year medical student project, paper submitted for publication.
- 2. **Shivani Subhedar (2018-2019):** Mentored UCSF Global Health Master’s student for capstone project.
- 3. **Amy Chen (2018-2019):** UCSF anesthesiology resident examining factors contributing to positive medical student clerkship in anesthesia.
- 4. **Laurence Henson (2019-2020):** Supervisor for UCSF Master’s in Epidemiology student, project on health outcomes and gentrification.
- 5. **Sofia Lochner (2021-2022):** Supervisor for UCSF Global Health Master’s project, examining National Surgical Obstetric and Anesthesia Plans.
- 6. **Lauren Locke (2022-2023):** Supervisor for residency Global Health track project on factors associated with anesthesia residents working with rural and underserved populations.

PAPERS REVIEWED

Journal	Manuscripts Reviewed Year	
Respiratory Care	1	2025
European Journal of Medical Research	1	2025
JAMA	1	2025
Journal of Neurosurgical Anesthesia	1	2024
Canadian Journal of Anesthesia	1	2022
	1	2023
	1	2024
	1	2025
Anesthesia & Analgesia	1	2023
International Journal of Health Policy and Management**	1	2022
Journal of Medical Education & Curricular Development	1	2022
British Journal of Anesthesia	2	2022
BMC Health Services Research	1	2022
Anesthesia Critical Care & Pain Medicine	1	2020
Advances in Medical Education and Practice	1	2020
Methodological Innovations	1	2019